



AI is managing invasive species

Canada is deploying artificial intelligence to help environmental managers manage aquatic invasive species. <https://digit.in/jun18-B41-1>



AI, future-seeing crystal ball

Researchers find that applying AI to data obtained from wearables can predict future health problems. <https://digit.in/jun18-B41-2>

as mice. If you have been interacting with the technology news chatbots of late, you will be well aware of a raging controversy on the ethics of using neural implants for competitive esports. We are in the process of developing implants for controlling computers, smartphones and smart devices. So far, this has been a very niche market, with few willing to undergo the surgery involved. However, with the reducing cost of the implants, along with the ease of use it promises, we forecast a healthy market for such devices over the course of the next ten years.

All the implants we have built so far, provide only a glimpse into the true potential of what these devices can really do. At most, the devices interface with about 3,000 neurons. There are hundred billion neurons in the human brain. As the neural implants so far have demonstrated, we do not need to directly interface with all the neurons in the brain. Somewhere between 10,000 to 100,000 neurons should allow us to do a lot more though, in the near future. However, before building better interfaces for the brain, we need to first figure out how the brain is wired, and then how all of it works together. Fortunately, half the work is already done, quite a bit ahead of schedule.

THE HUMAN CONNECTOME

In the 1970s, researchers started creating a wiring diagram of all the 302 neurons in the brain of a creature known as *Caenorhabditis elegans*, a nematode or a roundworm. 700 connections in the brain of the tiny, transparent creature was mapped. This was the first open source virtual worm brain, that allowed a number of researchers to carry out experiments with a simulated brain. The entire organism could be studied in a virtual environment. Behaviors such as finding food or mates, or avoiding toxins and predators were all seen in the virtual worm. The rather simple virtual brain was even taught to do tricks - such as balance a tiny pole on its tail. A real worm would respond in

a nearly identical fashion. This complete wiring diagram of the neurons, is known as a connectome.

As recently as 10 years ago, this nematode was the only organism with a mapped connectome. Since then, we have realised the connectomes of a fruit fly, a bat, a parrot, a dog, a monkey, and possibly the most important creature of them all.

In the year 2025, the Human Connectome Project successfully managed to create a connectome of an entire human brain. This was a mammoth undertaking, considering the sheer number of neurons in the human brain, and the seven hundred trillion synaptic connections. It was made possible because of advances in two fields, non invasive high resolution imaging, and artificial intelligence. The raw data was captured using a combination of magnetoencephalography, electroencephalography and magnetic resonance imaging.

Most of the work was done by an artificial intelligence built for that purpose. It sorted through hundreds of thousands of image slices of a brain, and then figured out which neurons were connected to which other neurons. Now this particular connectome was captured at a particular point in time. Although the subject was dead, in the human body, there are continuous changes in the brain. As we go through the business of living, synapses may form or break off, and grow larger and smaller. We have realised a human connectome, and the entire database is available online for researchers and software developers around the world, to tinker around with. Just obtaining the wiring map was a mammoth task, but now we do not know what exactly to do with it. In fact, we do not fully understand the connectome of the other, simpler animals as well. We have only gathered the data, the work on processing it remains to be done, and that is what an army of artificial intelligences are currently tasked with. We will have a better understanding of how the brain truly works, once we have finished

understanding this single human connectome, trapped in a slice of time. Concurrently, there are efforts to progress our understanding of the brain in other areas as well. We have to create the synaptome of the human brain, which is the states of the neurons, and the specific neurotransmitters present at the synapses.

THE ROAD AHEAD

Currently, our implants have tiny coils in them that receive power through wireless transmission. Not too many people like having batteries embedded in their skulls. At times there are glitches, and the implants just stop working, even though they are designed for 30 year life spans. If this happens, it is required to replace the device, which is costly and risky. Our current research involves finding a method to interface with the brain in a non invasive manner. There has been little development in this direction. The neurons actually shift within the brain, as we go through the business of living life. The brain is, after all, an extremely soft, malleable jelly like substance. While the work in soft electronics over the past decade has been promising, it is nowhere near good enough to handle the sheer amount of electrodes we will need for a true neural lace. When we crossed the 1,000 neuron mark, it made a tremendous increase in the range of implants we could create. Such leaps are expected when we create implants that can interface with 10,000 or 1,000,000 neurons. Neuralink, which makes neural interfaces for treating some conditions, is aiming for the one million mark. Then there is the question of the bandwidth needed to pull out or put in all this information in real time, and the energy required to support that bandwidth. Further advancements in material sciences, nanotechnology and fabrication are needed to fully realise the neural lace. 

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